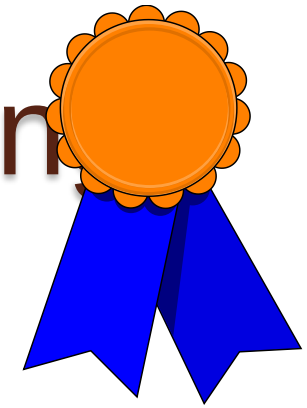
Learn C++ Programming



Lecture # 2

Objectives

- To learn fundamental problem solving techniques

- To learn how to design a program
- To learn to what is the C++ programming language and how to u

What is C++?

- C++ is a *programming language*.
- C++ ranks 4th in popularity according to 2017 IEEE spectrum Top Programming Language ranking. Learning C++ is a wise investment for all programmers.
- Creater: Bjrane Stroustrup

What is C++

- “C++ is a statically-typed, free-form, (usually) compiled, multi-paradigm, intermediate-level general-purpose middle-level programming language.”
- In simple terms, C++ is a sophisticated, efficient and a general-purpose programming language based on C. It was developed by Bjarne Stroustrup in 1979.
- Many of today’s operating systems, system drivers, browsers and games use C++ as their core language. This makes C++ one of the most popular languages today.
- Since it is an enhanced/extended version of C programming language, C and C++ are often denoted together as C/C++.

Features of C++

C++ is fast

Since, C++ is an extended version of C, the C part of it is very low level.

This offers a huge boost in speed that high level languages like Python, Java don’t give you.

- **<u>C++ is statically typed</u>**

  C++ is a statically typed programming language.

  In simple terms, C++ doesn’t allow the compiler to make assumptions about the type of data e.g. 10 is different from “10” and you have to let C++ know which one you are talking about.

- **<u>C++ is a multi-paradigm programming language</u>**

  C++ supports at least 7 different styles of programming and gives developers the freedom to choose one at their will.

  Unlike Java and Python, you don’t need to use objects to solve every task (if it’s not necessary).

  You can choose the programming style that fits your use case.

- **<u>Object oriented programming with C++</u>**

  Object oriented programming helps you solve a complex problem intuitively.

  With its use in C++, you are able to divide these complex problems into smaller sets by creating objects.

- **<u>Power of standard library (Standard template library - STL)</u>**

  The power of C++ extends with the use of standard libraries contained in it.

  These libraries contain efficient algorithms that you use extensively while coding.

  This saves ample amount of programming effort, which otherwise would have been wasted **reinventing the wheel**.

Reasons Why you should learn C++?

- **C++ is irreplaceable**

  With the use of C++ in development of modern games, operating systems, browsers, and much more, it is safe to say that C++ is irreplaceable.

  Many major applications likeAdobe Products like Photoshop, Illustrator, InDesign
- Amazon - one of the biggest e-commerce sites
- Autodesk products for Computer Aided Design
- Facebook - social networking site are

- **You learn the internal architecture of a computer**

  Since, C++ is a middle level language, you will write code that interacts directly with the internal hardware of the computer.

  You’ll learn how the computer memory really works, how information is stored in them, how you can retrieve them and so on.

  It is sure to expand your knowledge on the architecture of the computer.

- **Over 600,000 C++ repositories on Github**

  Github, the leading open source collaboration platform, has over 600,000 repositories for C++ alone.

  This metric itself proves the worth of C++ in the open source community as well.

  Be it gaming, graphics, windows applications, you can find tons of great open source projects extensively used today. And, you can always create your own.

- **60% StackOverflow Answer rate and active community**

  Likewise, with over 400,000 C++ questions asked on StackOverflow, the number one Q&A platform for developers, more than 60% questions have been answered.

  The number of questions asked and the percentage of them answered shows the interest and active support for C++ today.

  So, you can expect many great developers to help you solve real-life problems using C++.

- **C++ job opportunities and salary**

  C++ developers can expect an average of yearly $100,000 salary with over 7,700 jobs advertised every month.

  The requirement of jobs comes mostly from game development, rendering engines and the windows applications.

Compile and run C++ programming on your OS

- To run C++ Programming in Windows, you'd need to download Code::Blocks.
- Go to the binary release download page of Code:Blocks official site.

Windows XP / Vista / 7 / 8.x / 10:

File	Date	Download from
codeblocks-16.01-setup.exe	28 Jan 2016	**Sourceforge.net** or **FossHub**
codeblocks-16.01-setup-nonadmin.exe	28 Jan 2016	**Sourceforge.net** or **FossHub**
codeblocks-16.01-nosetup.zip	28 Jan 2016	**Sourceforge.net** or **FossHub**
codeblocks-16.01mingw-setup.exe	28 Jan 2016	**Sourceforge.net** or **FossHub**
codeblocks-16.01mingw-nosetup.zip	28 Jan 2016	**Sourceforge.net** or **FossHub**

- Under **Windows XP / Vista / 7 / 8.x / 10** section, click the link with **mingw-setup** highlighted row either from Sourceforge.net or FossHub.
- Open the Code::Blocks Setup file and follow the instructions (**Next** > **I agree** > **Next** > **Install**); you don’t need to change anything. This installs the Code::Blocks with gnu gcc compiler, which is the best compiler to start with for beginners.

```
#include <iostream>
using namespace std;
int main()
{
        cout<<"Hello World!";
        return 0;
  }
```